TECHNICAL SPECIFICATION – PALLET RACKING SYSTEM VerticalParts

Project: Vertical Storage Structure – Double Row
Client: VerticalParts – Warehouse Management
Data: February 01, 2026
Responsible: Gelson – Project and Process Management

This technical specification outlines the comprehensive design, implementation, and operational parameters for the new double-row pallet racking system at VerticalParts. The primary purpose of this document is to ensure a standardized approach to the installation, guaranteeing optimal space utilization, enhanced inventory accessibility, and improved operational efficiency within the warehouse. Key objectives include maximizing storage capacity, reducing material handling time, improving safety protocols, and establishing a scalable solution to support future growth. This system is designed to streamline VerticalParts' warehouse management, offering significant benefits through increased throughput and reduced operational costs.

CHAPTER 1

Executive Summary: System Overview

This project details the implementation of a complete Easytoque pallet rack system, designed to optimize VerticalParts' storage space. The solution features a robust and efficient configuration, distributed in two parallel rows to maximize storage capacity.

Model	A400xL230xP100
Levels	3 per module
Capacity	1,000 kg per level
Modules	19 (2 independent rows)

The Easytoque system was specifically chosen for its proven reliability, modular design, and high load-bearing capacity, which directly addresses VerticalParts' need for robust and scalable storage. This configuration, featuring two independent parallel rows, ensures maximum storage density without compromising accessibility, thereby achieving significant optimization benefits. By implementing a standardized model (A400xL230xP100) across 19 modules, VerticalParts will experience substantial efficiency gains through faster inventory placement and retrieval, reduced material handling errors, and streamlined workflow. The operational benefits include improved stock rotation, better space utilization by up to 30%, and a safer working environment due to the system's inherent stability and clear load specifications. Ultimately, this system is anticipated to boost overall warehouse throughput and reduce long-term operational costs by enhancing the speed and accuracy of all storage-related processes.

System Composition: General Configuration

The system consists of two independent rows, optimizing space utilization and allowing flexibility in warehouse organization. The table below details the module distribution for each row.

Description	Left Row	Right Row	Total
Initial Modules	1 unit	1 unit	2
Complementary Modules	8 units	9 units	17
Total Modules	9 modules	10 modules	19
Approximate Length	20,70 m	23,00 m	–

This configuration, featuring two distinct and independent rows, was specifically chosen to provide maximum operational flexibility and superior space optimization within the VerticalParts warehouse. By separating the storage into two rows (a left row of 9 modules and a right row of 10 modules), the system minimizes potential bottlenecks and allows for simultaneous operations, such as receiving goods in one aisle while picking in another, without interference. This independence significantly enhances the efficiency of material flow and accessibility to inventory. The differing lengths of the rows (20.70m and 23.00m respectively) were determined by the specific dimensions and architectural nuances of the warehouse, ensuring every available square meter is utilized effectively. This tailored distribution not only maximizes the storage capacity but also adapts to potential future expansion or reconfigurations more easily, offering long-term scalability and an optimized layout that directly addresses VerticalParts' unique storage requirements.

CHAPTER 2

Detailed Module Specification

Each module of the Easytoque system has been selected to ensure maximum safety and efficiency in storage, with technical specifications that meet the demands for heavy loads and high volume.

- **Model:** Initial and Complementary Pallet Rack A400xL230xP100 – 3 Levels
- **Height (A):** 4.000 mm (4.00 meters)
- **Clearance Width (L):** 2.300 mm (2.30 meters)
- **Depth (P):** 1.000 mm (1.00 meter)
- **Capacity per level:** 1.000 kg
- **Number of levels:** 3 levels
- **Total capacity per module:** 3.000 kg
- **Initial Module:** Independent structure with 2 vertical uprights
- **Complementary Module:** Shares 1 upright with the previous module

The structural design of each Easytoque module is meticulously engineered for robust performance and safety. The specified dimensions (4.00 meters height, 2.30 meters clearance width, and 1.00 meter depth) provide ample storage volume while maintaining optimal accessibility for forklifts and personnel. The independent structure of the initial modules, featuring two vertical uprights, establishes a strong foundational base, ensuring stability. Complementary modules then efficiently expand this system by sharing a single upright with the preceding module, a design choice that optimizes material usage without compromising structural integrity or load distribution. With each level capable of supporting 1,000 kg, and a total capacity of 3,000 kg per 3-level module, these specifications were selected to handle heavy loads typical in warehouse environments, minimizing the risk of structural failure and ensuring the safety of stored goods and operators. This high capacity per level also allows for versatile storage of various palletized items. Durability is a key consideration, with materials chosen to withstand continuous heavy use and resist wear and tear, ensuring a long operational lifespan. Furthermore, the modular design simplifies maintenance; individual components can be inspected and replaced if necessary, ensuring ongoing reliability and minimal downtime for the overall storage system.

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Component Quantity: Uprights

The vertical structure of the pallet racks is composed of robust uprights, ensuring the support and stability of the entire system. The distribution was calculated for each row, ensuring storage integrity.

Component	Left Row	Right Row
Uprights 4.00 m	10 units	11 units
Total Overall	21 units	

The uprights, each measuring 4.00 meters in height, are the foundational vertical elements that dictate the overall rack height and provide the primary support for the stored goods. Their robust construction is essential for handling the significant load-bearing capacity of 1,000 kg per level, culminating in a total capacity of 3,000 kg per three-level module. The distribution of 10 units for the Left Row and 11 units for the Right Row is meticulously calculated based on the modular design of the Easytoque system. Each independent "Initial Module" requires two uprights to establish its standalone structure, while subsequent "Complementary Modules" efficiently share one upright with the preceding module. This means the Left Row consists of one initial module and nine complementary modules (2 + (9-1)*1 = 10, or 1 initial module + 8 complementary modules), providing sufficient support for a total of 9 modules. Similarly, the Right Row's 11 uprights accommodate one initial module and 10 complementary modules (2 + (10-1)*1 = 11, or 1 initial module + 9 complementary modules), supporting a total of 10 modules. This optimized distribution ensures that each module, regardless of its position in the row, benefits from the necessary structural integrity to safely support its full load. Safety margins are inherently built into this design, guaranteeing stability and minimizing the risk of deflection or collapse under maximum operational load, thereby safeguarding both stored inventory and personnel.

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Component Quantity: Beams

The horizontal beams are crucial elements for forming the storage levels, directly supporting the pallet load. Their quantity was dimensioned to accommodate the 3 levels of each module.

Component	Left Row	Right Row
Beams 1,000 kg	54 units	60 units
Total Overall	114 units	

These horizontal beams, specified for a load capacity of 1,000 kg, are vital for creating stable storage levels within the pallet racking system. Each level is designed to support significant loads, ensuring safe and efficient storage of goods. The beams are typically manufactured from high-strength structural steel, providing exceptional durability and resistance to deformation under continuous heavy use. Their robust construction adheres to stringent safety standards, guaranteeing longevity and structural integrity.

The quantity of beams was meticulously calculated based on the system's modular design and the requirement for three storage levels per module. Considering that each module requires two beams per level (front and back) to securely support pallets, the calculation is as follows:

- **Left Row:** Comprising 9 modules, each with 3 levels, requires (9 modules * 3 levels * 2 beams/level) = 54 beams.
- **Right Row:** Comprising 10 modules, each with 3 levels, requires (10 modules * 3 levels * 2 beams/level) = 60 beams.

This precise distribution ensures optimal load distribution across all levels, allowing each beam to consistently support its 1,000 kg capacity. The quality of the material and the engineering behind these specifications are critical for preventing structural fatigue and ensuring the long-term reliability and safety of the entire storage solution.

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CHAPTER 4

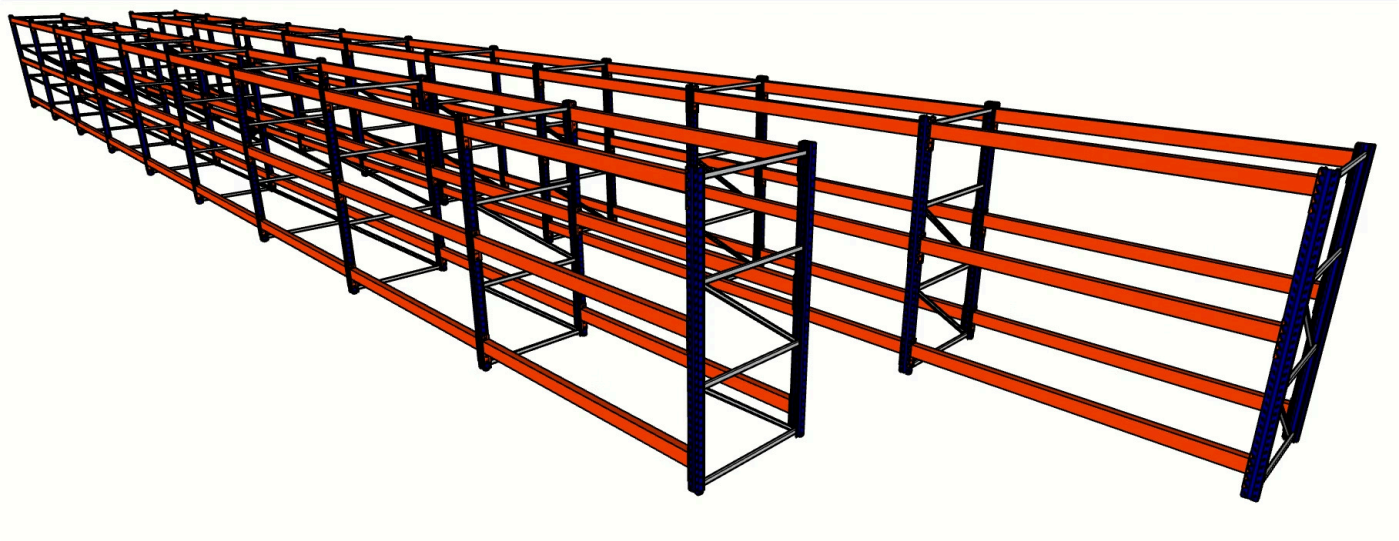
Layout and Spatial Distribution

The proposed "L" arrangement maximizes the use of the warehouse space, creating a functional central corridor for equipment traffic and cargo movement. The configuration ensures accessibility and operational safety.

This "L" shaped layout is strategically chosen to optimize the warehouse footprint, particularly in facilities with specific architectural constraints or to delineate distinct operational zones. It provides an efficient flow for goods receiving, storage, and dispatch, significantly reducing transit times and improving overall material handling efficiency. This arrangement inherently streamlines internal logistics, defining clear primary paths for both mechanized equipment and personnel, which minimizes potential bottlenecks and enhances operational throughput.

A critical component of this design is the central corridor. At a minimum width of 3.00 meters, it is dimensioned to accommodate the safe and simultaneous passage of various types of industrial equipment, such as counterbalance forklifts and reach trucks, even when carrying wide loads. This width provides ample maneuvering space, crucial for preventing collisions and ensuring clear sightlines for operators, thereby mitigating risks and adhering to stringent safety regulations for aisle clearances. Furthermore, it ensures unhindered accessibility for emergency services and facilitates routine maintenance and inventory management tasks.

The combination of precise row depths and an optimized corridor width is essential for achieving high-density storage without compromising operational fluidity or safety. The 1.00-meter depth per row allows for efficient pallet storage while maintaining structural integrity. The overall height of 4.00 meters maximizes vertical storage capacity, making full use of the available air space to store more goods per square meter of floor space. This holistic approach to spatial distribution not only enhances efficiency but also provides a safer, more organized working environment.



Left Row: 9 continuous modules, ~20.70 m

Right Row: 10 continuous modules, ~23.00 m

Central Corridor: Minimum width 3.00 m

Depth per Row: 1.00 m

Total Height: 4.00 m

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CHAPTER 4

Visual Representations

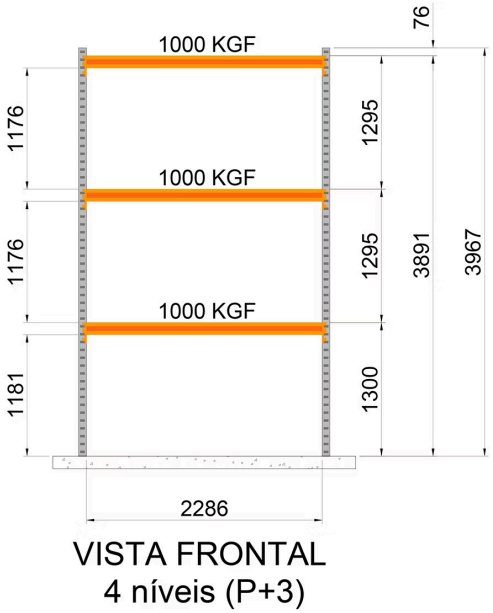
For a complete understanding of the project, various visual perspectives of the Easytoque pallet rack system are provided, illustrating the robustness and functionality of the proposed solution.

Isometric View

Initial

Top View

Initial



Left Side View

Right Side View

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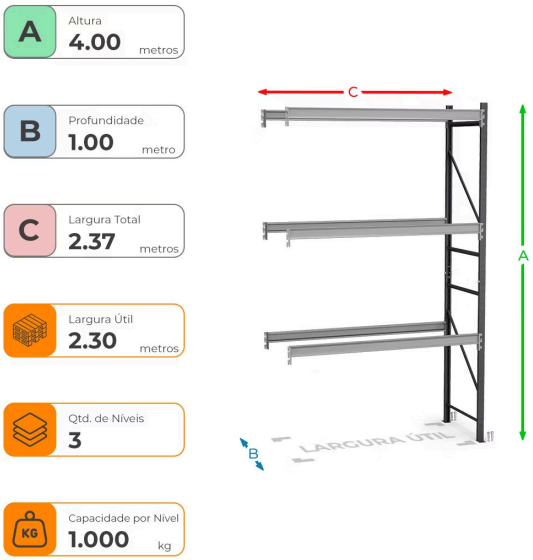
CHAPTER 4

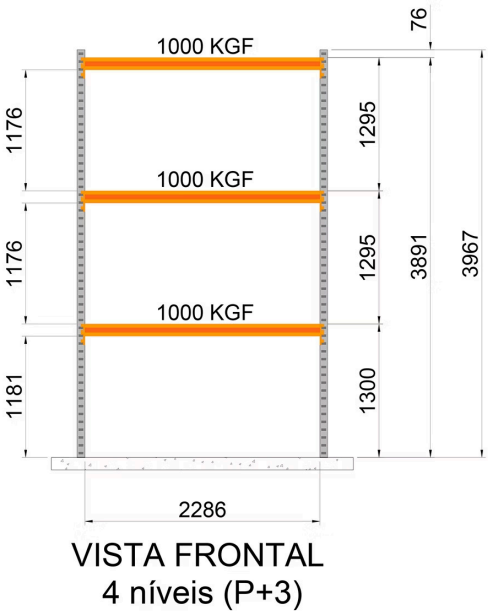
Visual Representations

For a complete understanding of the project, various visual perspectives of the Easytoque pallet rack system will be provided, including detailed views and product images, illustrating the robustness and functionality of the proposed solution.



Isometric View Complementary

Top View Complementary



Left Side View

Right Side View

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CHAPTER 5

Budget and Cost Analysis

The price composition details the investment required for the acquisition of the complete pallet racking system, considering the distinction between initial and complementary modules.

Item	Qty	Unit Value	Total Value
Initial Pallet Racking A400xL230xP100 – 3 levels	2	R$ 2.429,90	R$ 4.859,80
Complementary Pallet Racking A400xL230xP100 – 3 levels	17	R$ 1.699,90	R$ 28.898,30
Subtotal Complete System	**19**	–	**R$ 33.758,10**

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Comparative Analysis: Complete Sets vs. Loose Components

To optimize the investment, it is recommended to request comparative quotes considering two acquisition modalities: complete sets (initial and complementary) provided by the manufacturer Easytoque, and the purchase of individual components (uprights and beams) for customized assembly. The analysis should consider prices, delivery times, warranties, and installation costs.

Acquisition Considerations

For a detailed quotation from suppliers (national or international), it will be necessary to specify the following components:

- **21** Uprights of 4.00 m
- **114** Beams of 1,000 kg

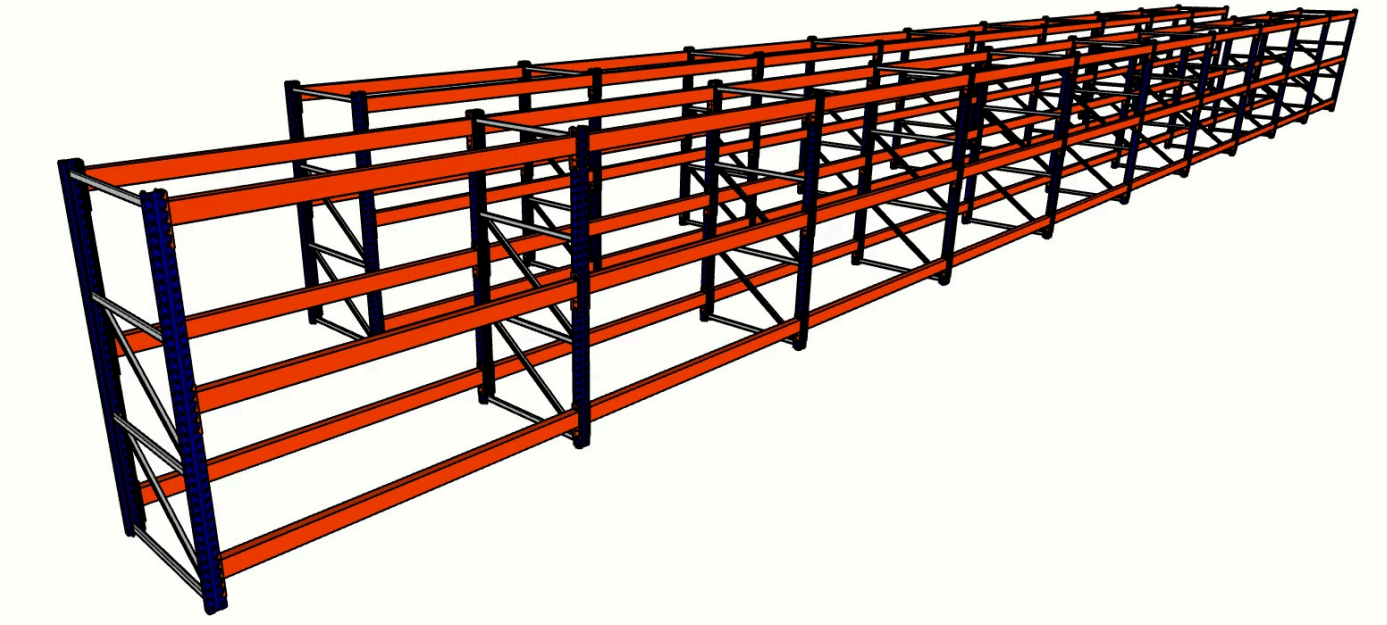
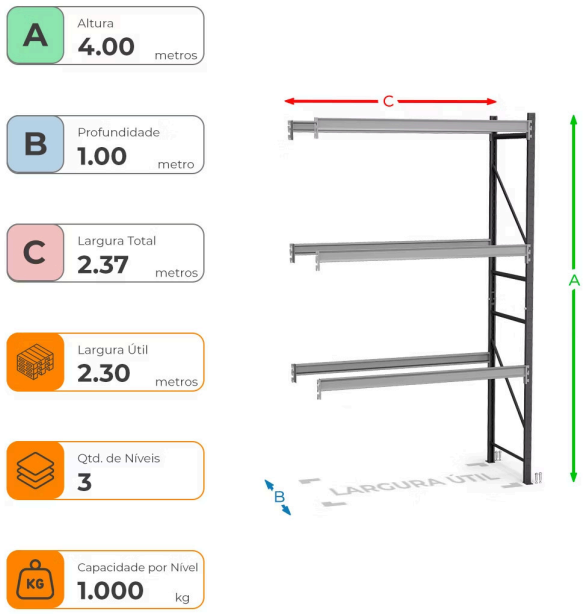


Image: Complementary Pallet Rack A400xL230xP100 - 3 Levels 1,000 Kg Per Level



Modular complementary pallet rack structure model A400xL230xP100, with 4.00 meters in height, 3 levels of horizontal storage with individual capacity of 1,000 kg per level. System designed for efficient vertical stock management. 17 of these complementary pieces with 4.00 meters in height will be needed to complete VerticalParts' storage project.

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Pallet Rack Starter Unit A400xL230xP100 - 3 Levels 1,000 Kg Per Level

Modular starter pallet rack structure, model A400xL230xP100, with 4.00 meters in height, 3 horizontal storage levels with an individual capacity of 1,000 kg per level. System designed for efficient vertical stock management. 02 of these Starter units, 4.00 meters high, will be needed to complete the VerticalParts storage project.

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